Short-Rate Models and Bond Pricing

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Spot Rate, Forward Rates, and the Zero-Coupon Yield Curve

The Zero-Coupon Bond: The Fundamental Building Block

Definition. A *zero-coupon bond* with maturity T pays \$1 at time T and nothing before.

$$P(t, T) \;=\; \text{price at time } t \leq T.$$

Risk-neutral pricing formula:

$$\boxed{P(t, T) \;=\; \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_s \, \mathrm{d}s} \;\middle|\; \mathcal{F}_t \right]}$$

- $e^{-\int_t^T r_s \, \mathrm{d}s}$: stochastic discount factor from t to T.
- r_s: the *instantaneous short rate* at time s.
- $P(t, t) = 1$ (a bond that matures now is worth its face value).
- $P(t, T) \in (0, 1]$ under positive interest rates.

Where Does This Formula Come From?

Start in discrete time. In a one-period world with constant rate r, the discounted price process is:

$$B_t^{-1} \cdot S_t \;=\; S_t \cdot \prod_k \frac{1}{1 + r\,\Delta t}.$$

Pass to continuous time. With a *constant* continuously-compounded rate r, the money-market account is $B_t = e^{rt}$, so:

$$B_t^{-1} S_t \;=\; S_t\, e^{-rT}.$$

Problem. A single constant rate over the whole period is unrealistic:

- The economy faces different inflation pressures over time.
- The time-value of money is constantly changing.
- $\Rightarrow$ We need to *generalize* the discounted price process.

The Instantaneous Short Rate and Money-Market Account

Key idea. Let the interval $[0, T]$ be sliced into infinitesimally small pieces of length $\Delta t \to 0$. In each piece $[t, t + \Delta t]$, the rate is r_t.

Letting $\Delta t \to 0$, we obtain a *single rate for an infinitesimally short period*: the **instantaneous short rate** r_t.

The money-market account. Investing \$1 at time 0 and continuously rolling over at r_t:

$$B_t = \exp\left(\int_0^t r_s \, \mathrm{d}s\right), \qquad \frac{\mathrm{d}B_t}{B_t} = r_t \, \mathrm{d}t.$$

The discounted price process is now generalized to constantly changing rates:

$$B_t^{-1} S_t = S_t \cdot \exp\left(-\int_0^t r_s \, \mathrm{d}s\right).$$

Bond Yield and the Yield Curve

Bond price from yield. We write the zero-coupon bond price as:

$$P(t, T) \;=\; e^{-y(t,T)\,(T-t)},$$

where $y(t, T)$ is the **continuously-compounded zero-coupon yield**.

Solving for the yield:

$$y(t, T) \;=\; -\frac{1}{T-t}\,\ln P(t, T).$$

$y(t, T)$ is the constant rate that would give the *same* bond price as the full stochastic discounting — a summary statistic of the bond.

Since bonds have many maturities ($T = 1, 2, 5, 30, \ldots$ years), the function $T \mapsto y(t, T)$ traces the **zero-coupon yield curve** at time t.

Forward Rates: The Marginal Rate at a Future Date

The instantaneous forward rate $f(t, T)$ is the marginal rate the market implies for borrowing over an infinitesimally short interval around a *future* date T, as seen from time t:

$$f(t, T) \;:=\; -\frac{\partial}{\partial T} \ln P(t, T).$$

Intuition. $f(t, T)\,\Delta T \approx$ rate paid (locked in today at t) for borrowing from T to $T + \Delta T$.

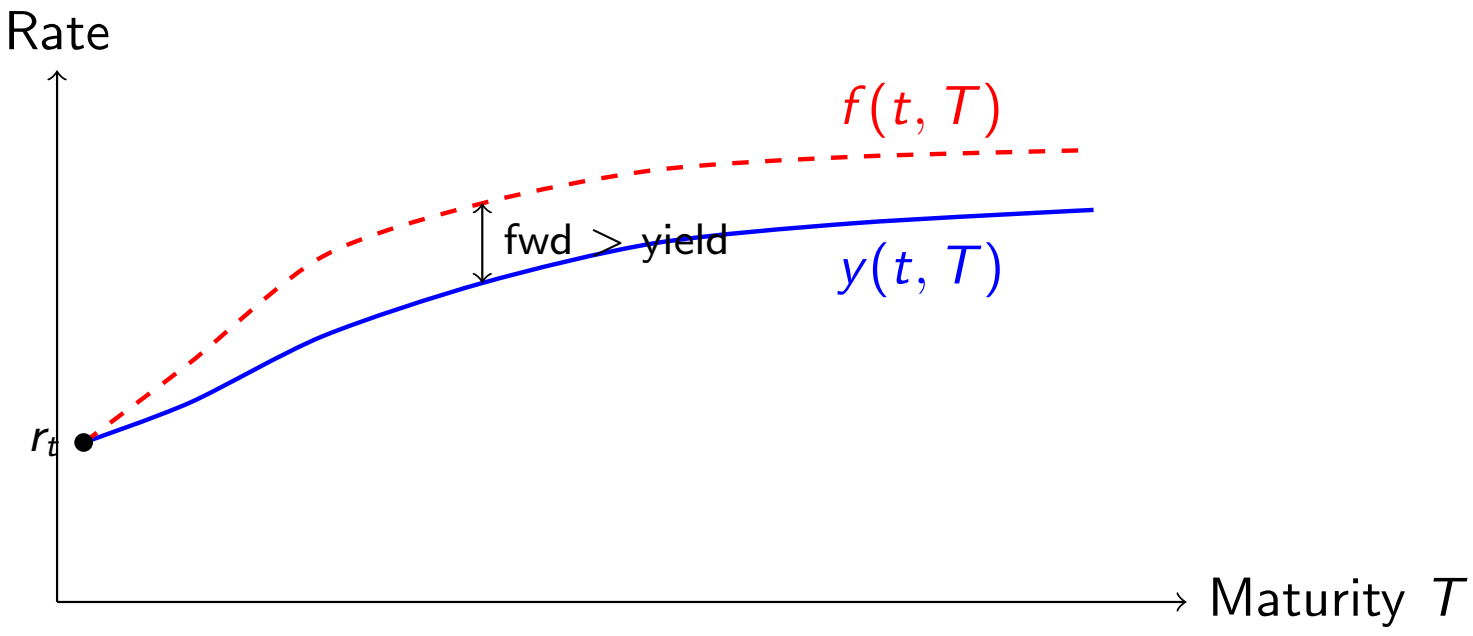
Bootstrapping forward rates from bond prices. Integrating $f(t, T) = -\partial_T \ln P(t, T)$ over $[t, T]$ and using $P(t, t) = 1$:

$$\boxed{P(t, T) \;=\; \exp\left(-\int_t^T f(t, u)\,\mathrm{d}u\right).}$$

Bond price = exponential of the *area under the forward curve* from t to T.

Yield as average of forward rates:

The Yield Curve and Forward Curve Together



Key facts:

- Both curves pass through r_t as $T \to t$ (the short rate).
- $f(t, T) > y(t, T)$ when the yield curve is upward-sloping.
- $r_t = \lim_{T \downarrow t} y(t, T) = f(t, t)$: the short rate is the *instantaneous* yield.

The Key Relationship: Discount Rates and Forward Rates

Under $\mathbb{Q}$, combining the two representations:

Discounted price process

$$B_t^{-1} S_t = S_t \, e^{-\int_0^t r_s \, \mathrm{d}s}$$

(known up to time t)

Bond price

$$P(t, T) = e^{-\int_t^T f(t,u) \, \mathrm{d}u}$$

(instantaneous forward rates)

We establish the fundamental relationship:

$$P(t, T) \;=\; \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_t^T r_s \, \mathrm{d}s} \,\Big|\, \mathcal{F}_t \Big] \;=\; e^{-\int_t^T f(t,u) \, \mathrm{d}u}.$$

Differentiating with respect to T:

$$f(t, T) \;=\; -\frac{\partial}{\partial T} \ln P(t, T).$$

This allows *bootstrapping* $f(t, T)$ from market bond prices at time t.

Why Do We Need to Model r_t?

In a simple market model with *constant* rates, discounting is trivial. But:

- The economy is always subject to different rates and inflationary pressures.
- The time-value of money is *constantly changing*.

Advantages of the instantaneous short rate r_t:

- Flexible — accounts for a changing interest rate environment.
- Allows exact discounting at each instant.

New problem this creates:

- The risk from rates is now something we need to *model and predict* to ensure fair, no-arbitrage pricing.
- Under $\mathbb{P}$: r_t follows dynamics calibrated to historical data.
- Under $\mathbb{Q}$: r_t follows risk-neutral dynamics used for pricing.

$\Rightarrow$ We need a stochastic model for r_t and a way to change measure.

Risk-Neutral Dynamics Under $\mathbb{Q}$: The CIR Model

CIR Under the Physical Measure $\mathbb{P}$

Assumption: We work on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$.

The **Cox–Ingersoll–Ross (CIR)** model specifies the short rate under $\mathbb{P}$:

$$\mathrm{d}r_t \;=\; \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\,\mathrm{d}t \;+\; \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{P}}, \qquad r_0 > 0.$$

- $\kappa_{\mathbb{P}} > 0$: speed of mean reversion.
- $\theta_{\mathbb{P}} > 0$: long-run level under $\mathbb{P}$ (physical equilibrium rate).
- $\sigma > 0$: volatility parameter.
- $\sigma\sqrt{r_t}$: *state-dependent* volatility — higher when rates are higher.

Feller condition (for strict positivity):

$$2\kappa_{\mathbb{P}}\theta_{\mathbb{P}} \;\geq\; \sigma^2 \quad \Rightarrow \quad r_t > 0 \;\text{ a.s. for all } t.$$

Why Can't We Price Directly Under $\mathbb{P}$?

No-arbitrage requires a risk-neutral measure $\mathbb{Q}$.

- Under $\mathbb{P}$: expectations reflect *real-world* probabilities.
- For pricing: we need $\mathbb{Q}$ under which every discounted asset is a *martingale*.
- $\mathbb{P}$ and $\mathbb{Q}$ are **equivalent** (agree on zero-probability events) but assign different probabilities to non-null events.

Radon-Nikodym derivative (the "exchange rate" between measures):

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}\left[X \cdot \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right].$$

The bridge between $\mathbb{P}$ and $\mathbb{Q}$ for Brownian-driven models is **Girsanov's theorem**.

Girsanov's Theorem

Setup. Let $(\lambda_t)_{t\geq 0}$ be an adapted process (the *market price of risk*). Define:

$$Z_T = \exp\left(-\int_0^T \lambda_s \, \mathrm{d}W_s^{\mathbb{P}} - \frac{1}{2}\int_0^T \lambda_s^2 \, \mathrm{d}s\right).$$

Set $\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T$.

Girsanov's theorem: Under $\mathbb{Q}$, the process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + \int_0^t \lambda_s \, \mathrm{d}s$$

is a **standard Brownian motion**.

Equivalently: $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \, \mathrm{d}t$.

Intuition. We add a deterministic drift $\lambda_t \, \mathrm{d}t$ to the $\mathbb{P}$-Brownian motion. The "new" randomness is now centred under $\mathbb{Q}$.

From $\mathbb{P}$ to $\mathbb{Q}$: Rewriting the CIR SDE

Under $\mathbb{P}$:

$$\mathrm{d}r_t = \kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{P}}.$$

Substituting $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t\,\mathrm{d}t$:

$$\mathrm{d}r_t = \underbrace{\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\,\mathrm{d}t}_{\text{physical drift}} + \sigma\sqrt{r_t}\underbrace{(\mathrm{d}W_t^{\mathbb{Q}} - \lambda_t\,\mathrm{d}t)}_{\mathrm{d}W_t^{\mathbb{P}}}$$

Expanding:

$$\mathrm{d}r_t = \Big[\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t) - \sigma\sqrt{r_t}\,\lambda_t\Big]\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Goal: choose λ_t so that the $\mathbb{Q}$-drift is again *affine in* r_t:

$$\mathrm{d}r_t = \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Note: Girsanov changes only the *drift* — the diffusion coefficient $\sigma\sqrt{r_t}$ is **the same under both measures**.

The Required Form of λ_t

Matching the drift. We need:

$$\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t) - \sigma\sqrt{r_t}\,\lambda_t \;=\; \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} - \kappa_{\mathbb{Q}} r_t.$$

For this to hold for all $r_t > 0$ with the CIR structure preserved, set:

$$\lambda_t \;=\; \eta\sqrt{r_t}, \qquad \eta \in \mathbb{R}.$$

This gives the **risk-neutral parameters**:

$$\boxed{\kappa_{\mathbb{Q}} \;=\; \kappa_{\mathbb{P}} + \sigma\eta, \qquad \kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} \;=\; \kappa_{\mathbb{P}}\theta_{\mathbb{P}}.}$$

- $\lambda_t = \eta\sqrt{r_t}$: risk premium is proportional to $\sqrt{r_t}$ — higher when rates are high.
- $\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} = \kappa_{\mathbb{P}}\theta_{\mathbb{P}}$: product of speed and long-run level is *preserved* under the measure change.
- Diffusion coefficient $\sigma\sqrt{r_t}$ is unchanged under $\mathbb{Q}$.

The Feller Condition: Keeping Rates Positive

CIR under $\mathbb{Q}$:

$$\mathrm{d}r_t = \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Behavior near $r_t = 0$:

- Diffusion $\sigma\sqrt{r_t} \approx 0$: very little noise.
- Drift $\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} > 0$: pushes rate *upward*.
- When $r_t < \theta_{\mathbb{Q}}$: upward pull; when $r_t > \theta_{\mathbb{Q}}$: downward pull.

Feller condition:

$$2\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} \;\geq\; \sigma^2.$$

Guarantees: upward drift at zero is strong enough to prevent r_t from reaching 0.

$2\kappa\theta > \sigma^2$	Strict Feller	$r_t > 0$ a.s. (zero *inaccessible*)
$2\kappa\theta = \sigma^2$	Boundary	r_t touches 0 but reflects instantly
$2\kappa\theta < \sigma^2$	Violated	r_t can spend time at 0

Note: violating Feller does **not** destroy the strong solution — $r_t \geq 0$ a.s.

The Novikov Condition

Why we need it. For $Z_T = \exp\big(-\int_0^T \lambda_s \, \mathrm{d}W_s^{\mathbb{P}} - \frac{1}{2}\int_0^T \lambda_s^2 \, \mathrm{d}s\big)$ to define a valid probability measure $\mathbb{Q}$, Z_T must be a *true martingale* (not merely local).

Novikov condition (sufficient):

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \lambda_t^2 \, \mathrm{d}t\right)\right] < \infty.$$

Verification for $\lambda_t = \eta\sqrt{r_t}$.

$$\frac{1}{2}\int_0^T \lambda_t^2 \, \mathrm{d}t \;=\; \frac{\eta^2}{2}\int_0^T r_t \, \mathrm{d}t.$$

We need: $\mathbb{E}^{\mathbb{P}}\big[\exp\big(\frac{\eta^2}{2}\int_0^T r_t \, \mathrm{d}t\big)\big] < \infty.$

Since CIR is an **affine process**, its moment generating function for $\int_0^T r_t \, \mathrm{d}t$ is exponential-affine in r_0 and *finite on any finite horizon* T.

$\Rightarrow$ **Novikov condition is satisfied** for any finite T and any $\eta \in \mathbb{R}$.

Why $\lambda_t = \eta\sqrt{r_t}$ Is the Natural Choice

The problem with other choices of λ_t.

Suppose instead $\lambda_t = \zeta/\sqrt{r_t}$ (e.g. if we tried a constant-volatility correction):

- As $r_t \to 0$: $\lambda_t \to \infty$ — the integrand blows up.
- $\int_0^T \lambda_t^2 \, \mathrm{d}t = \zeta^2 \int_0^T r_t^{-1} \, \mathrm{d}t$ can be **unbounded**.
- Novikov condition *may fail* — $\mathbb{Q}$ would not be a valid probability measure.

Why $\lambda_t = \eta\sqrt{r_t}$ works:

- As $r_t \to 0$: $\lambda_t \to 0$ — the market price of risk vanishes.
- $\int_0^T \eta^2 r_t \, \mathrm{d}t$ is bounded (CIR is mean-reverting and non-negative).
- Novikov condition holds on any finite $[0, T]$.
- *CIR structure is preserved* under $\mathbb{Q}$.

Economic interpretation: risk compensation $\propto \sqrt{r_t}$ — higher when rates are elevated, near zero in a liquidity-trap environment.

CIR Preserved Under $\mathbb{Q}$: Full Derivation

Starting from the $\mathbb{P}$-dynamics and substituting $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \eta\sqrt{r_t}\,\mathrm{d}t$:

$$\begin{aligned}
\mathrm{d}r_t &= \left[\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t) - \sigma\sqrt{r_t}\cdot\eta\sqrt{r_t}\right]\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}} \\
&= \left[\kappa_{\mathbb{P}}\theta_{\mathbb{P}} - \kappa_{\mathbb{P}}r_t - \sigma\eta r_t\right]\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}} \\
&= \left[\kappa_{\mathbb{P}}\theta_{\mathbb{P}} - (\kappa_{\mathbb{P}} + \sigma\eta)\,r_t\right]\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}.
\end{aligned}$$

Set $\kappa_{\mathbb{Q}} := \kappa_{\mathbb{P}} + \sigma\eta$ and $\theta_{\mathbb{Q}} := \dfrac{\kappa_{\mathbb{P}}\theta_{\mathbb{P}}}{\kappa_{\mathbb{Q}}}$. Then:

$$\boxed{\mathrm{d}r_t \;=\; \kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

The **CIR structure is fully preserved** under $\mathbb{Q}$:

- Same diffusion coefficient $\sigma\sqrt{r_t}$ (Girsanov only changes the drift).
- Same affine structure $\Rightarrow$ closed-form bond pricing formulas.

Summary: $\mathbb{P}$–$\mathbb{Q}$ Relationship in the CIR Model

	Under $\mathbb{P}$	Under $\mathbb{Q}$
SDE	$\kappa_{\mathbb{P}}(\theta_{\mathbb{P}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{P}}$	$\kappa_{\mathbb{Q}}(\theta_{\mathbb{Q}} - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}}$
Speed	$\kappa_{\mathbb{P}}$	$\kappa_{\mathbb{Q}} = \kappa_{\mathbb{P}} + \sigma\eta$
Long-run	$\theta_{\mathbb{P}}$	$\theta_{\mathbb{Q}} = \kappa_{\mathbb{P}}\theta_{\mathbb{P}}/\kappa_{\mathbb{Q}}$
Diffusion	$\sigma\sqrt{r_t}$	$\sigma\sqrt{r_t}$ (unchanged)

Key invariants:

$$\kappa_{\mathbb{Q}}\theta_{\mathbb{Q}} \;=\; \kappa_{\mathbb{P}}\theta_{\mathbb{P}} \qquad \text{(Feller constant preserved)}$$

σ is the same under both measures.

Next: Use the $\mathbb{Q}$-dynamics to show the discounted bond price is a $\mathbb{Q}$-martingale, then derive the log-affine CIR bond pricing formula via Feynman-Kac.